FOR RASPBERRY PI 4 (Raspbian Buster, Python 3.7), Follow step by step

UPDATE

//Solves problems when doing “sudo apt-get install”

```
sudo apt update && sudo apt full-upgrade -y
sudo apt update && sudo apt full-upgrade -y --fix-missing
```

PIP3

```
sudo apt-get install python3-pip
pip3 install --upgrade pip
```

PYGAME

```
sudo pip3 install pygame
```

DJANGO

```
sudo pip3 install django
```

MYSQL CLIENT

```
CHECK GUIDE 3 MySQL to Rpi
```

CAMERA

```
sudo apt-get install pciutils
sudo setpci -s 01:00.0 0xD4.B=0x41
```

DHT22 SENSOR

sudo pip3 install Adafruit_DHT

FC-28 SOIL MOISTURE SENSOR

Enabling the SPI Interface with the Raspberry Pi

Follow these steps to enable the SPI interface with the Raspberry Pi:

1. Start by launching the terminal and typing in the below command:
   sudo raspi-config
2. Navigate to the interfacing options.
3. Enable the SPI interface.
4. Reboot your Raspberry Pi.

Installing the spidev Library

The spidev library will help with reading the sensor values over the SPI interface.
Use the following commands to install the spidev library:
sudo apt-get install git python-dev
git clone git://github.com/doceme/py-spidev
cd py-spidev/
sudo python setup.py install

Installing the numpy Library

The numpy library is also required in order to convert the output values to a percentage. The output values we receive from the MCP3008 IC will be in the 0-0123 range mentioned earlier. We'll then map these values to 0-100 for a percentage.
Install the numpy module with the following command:
sudo apt-get install python3-numpy

IMAGE PROCESSING

CHECK GUIDE INSIDE IMAGE PROCESSING FOLDER